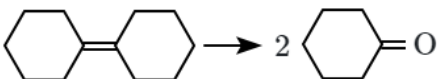
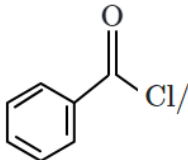
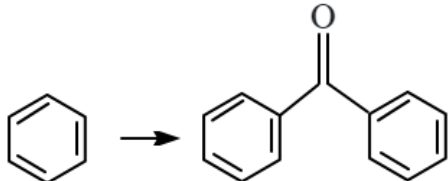
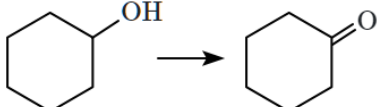
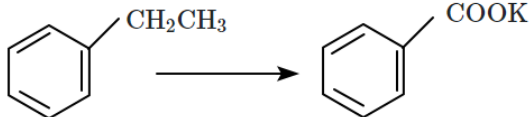


Q1 Match List I with List II.

NEET - 2024

List I (Reaction)	List II (Reagents/ Condition)
A. 	I.  Anhyd. AlCl_3
B. 	II. $\text{CrO}_3/\text{CrO}_3$
C. 	III. $\text{KMnO}_4/\text{KOH}, \Delta$ $\text{KMnO}_4/\text{KOH}, \Delta$
D. 	IV. (i) O_3 (ii) $\text{Zn}-\text{H}_2\text{O}$ (i) O_3 (ii) $\text{Zn}-\text{H}_2\text{O}$

Choose the correct answer from the options given below :

1. A-III, B-I, C-II, D-IV A-III, B-I, C-II, D-IV
2. A-IV, B-I, C-II, D-III A-IV, B-I, C-II, D-III

3. A-I, B-IV, C-II, D-III A-I, B-IV, C-II, D-III

4. A-IV, B-I, C-III, D-II A-IV, B-I, C-III, D-II

Q2 Fehling solution 'A' is: **NEET - 2024**

1. Alkaline copper sulphate
2. Alkaline solution of sodium potassium tartrate (Rochelle's salt)
3. Aqueous sodium citrate
4. Aqueous copper sulphate

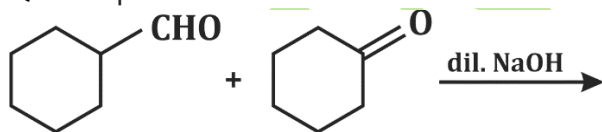
Q3. Given below are two statements: **NEET - 2024**

Statement I:	Benzoic acid produces effervescence on treatment with aq. NaHCO_3
Statement II:	The effervescence is due to the release of hydrogen gas.

In the light of the above statements, choose the not appropriate answer from the options give below:

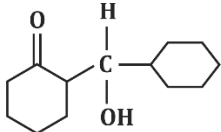
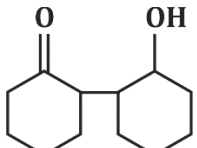
1. Both **Statement I** and **Statement II** are correct.
2. Both **Statement I** and **Statement II** are incorrect.
3. **Statement I** is correct but **Statement II** is incorrect.
4. **Statement I** is incorrect but **Statement II** is correct.

Q4. The product that cannot be formed in the following reaction is: **NEET - 2024**



1.	
2.	



3.	
4.	

Q5. Match List-I with List-II

NEET - 2024

	List-I (Reagent)		List-II (Name of the reaction)
A.	$\text{H}_2, \text{Nd}-\text{BaSO}_4, \text{H}_2, \text{Nd}-\text{BaSO}_4$	I.	Gattermann-koch reaction
B.	(i) $\text{CrO}_2\text{Cl}_2, \text{CS}_2$ (ii) H_2C (i) $\text{CrO}_2\text{Cl}_2, \text{CS}_2$ (ii) H_2C	II.	Reimer-Tiemann reaction
C.	$\text{CO}, \text{HCl}, \text{Anhyd. AlCl}_3 / \text{CuCl}$ $\text{CO}, \text{HCl}, \text{Anhyd. AlCl}_3 / \text{CuCl}$	III.	Etard reaction
D.	$\text{CHCl}_3, \text{NaOH}$ $\text{CHCl}_3, \text{NaOH}$	IV.	Rosenmund reduction

Choose the correct answer from the options given below:

1. A-II, B-III, C-I, D-IV

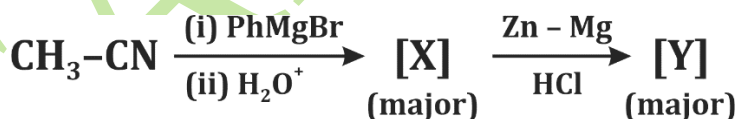
2. A-IV, B-III, C-I, D-II

3. A-I, B-III, C-IV, D-II

4. A-II, B-I, C-III, D-IV

Q6. Identify [X] and [Y] in given reaction:

NEET - 2024



1.	$[\text{X}] = \text{Ph} \begin{array}{c} \text{O} \\ \parallel \\ \text{---} \end{array}, [\text{Y}] = \text{Ph} \begin{array}{c} \text{OH} \\ \\ \text{---} \end{array}$
----	---



2.	$[X] = \text{Ph} - \text{CH}(\text{NH}_2) - \text{CH}_3, [Y] = \text{Ph} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$
3.	$[X] = \text{Ph} - \text{C}(=\text{O}) - \text{CH}_3, [Y] = \text{Ph} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$
4.	$[X] = \text{Ph} - \text{CH}(\text{NH}_2) - \text{CH}_3, [Y] = \text{Ph} - \text{CH}(\text{OH}) - \text{CH}_3$

Q7. Select the incorrect reaction among the following: NEET - 2024

1.	$\text{CH}_3\text{COCl} \xrightarrow{\text{H}_2\text{O}} \text{CH}_3\text{COOH} \quad \text{CH}_3\text{COCl} \xrightarrow{\Delta} \text{CH}_3\text{COOH}$
2.	Reaction scheme: Benzamide (a benzene ring with a CONH_2 group) reacts with (i) LiAlH_4 and (ii) H_2O to form benzoic acid (a benzene ring with a COOH group).
3.	$\text{CH}_3\text{CH}_2\text{OH} \xrightarrow{\text{(i) KMnO}_4/\text{OH}^-} \text{CH}_3\text{COOH} \quad \text{CH}_3\text{CH}_2\text{OH} \xrightarrow{\text{(ii) H}_3\text{O}^+} \text{CH}_3\text{CH}_2\text{COOH}$
4.	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow{\text{CrO}_3-\text{H}_2\text{SO}_4} \text{CH}_3\text{CH}_2\text{COOH}$

Q8. Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in the presence of calcium oxide is:

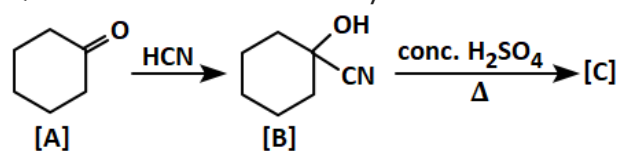
NEET - 2023

1. 18
2. 16
3. 32
4. 30

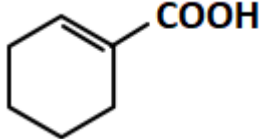
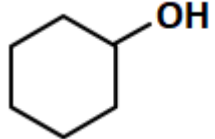
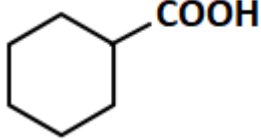
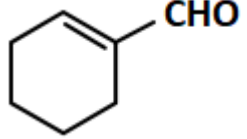


Q9. Given is a reaction for your reference:

NEET – 2023

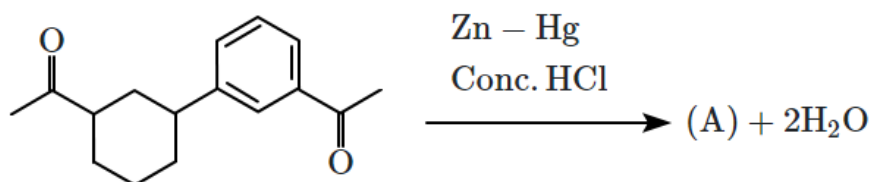


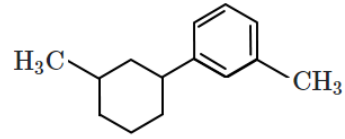
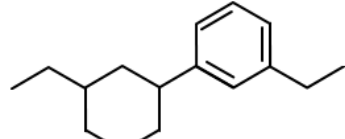
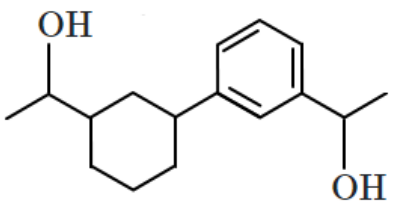
The final product [C] is:

1.		2.	
3.		4.	

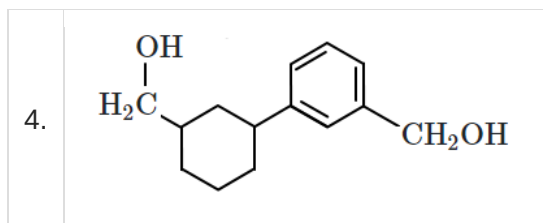
Q10. Identify the product (A) in the following reaction:

NEET – 2023

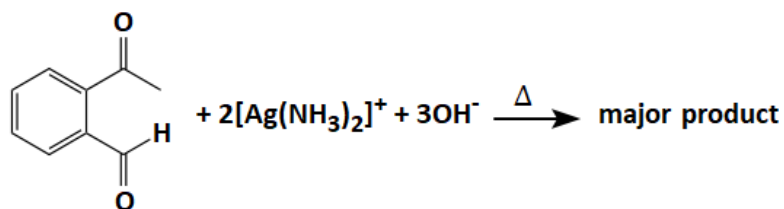


1.	
2.	
3.	





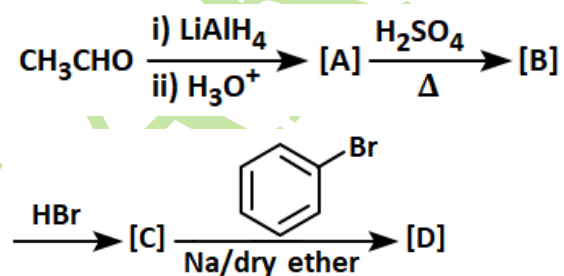
Q11. Identify the major product obtained in the following reaction: **NEET - 2023**



1.		2.	
3.		4.	

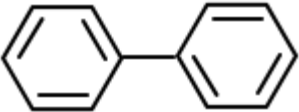
Q12. Identify the final product [D] obtained in the following sequence of reactions:

NEET - 2023



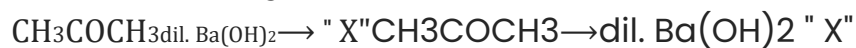
1.	$\text{HC}\equiv\text{C}^- \text{Na}^+ + \text{HC}\equiv\text{C}^- \text{Na}^+$	2.	
----	---	----	--



3.		4.	C_4H_1
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Q13 Consider the given reaction:

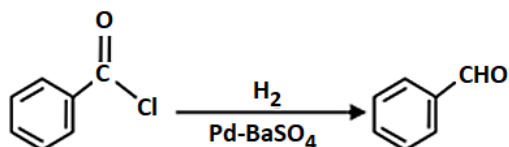
NEET - 2023



The functional groups present in the compound "X" are:

1. Ketone and double bond
2. Double bond and aldehyde
3. Alcohol and aldehyde
4. Alcohol and ketone

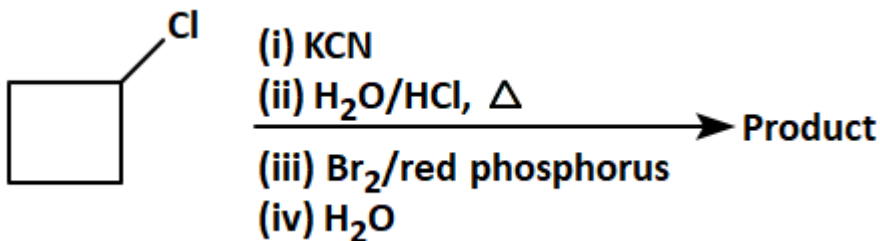
Q14 Mark the name of the reaction associated with the following conversion NEET - 2023

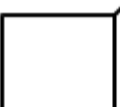


1. Stephen reaction
2. Gattermann-Koch reaction
3. Etard reaction
4. Rosenmund reaction

Q15. Identify the product in the following reaction:

NEET - 2023



1.		2.	
3.		4.	

Q16. Given below are two statements: **NEET - 2022**

Statement I	The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular masses because of weak molecular association in aldehydes and ketones due to dipole-dipole interactions.
Statement II	The boiling points of aldehydes and ketones are lower than the alcohols of similar molecular masses due to the absence of H-bonding.

In the light of the above statements, choose the most appropriate answer from the options given below:

1.	Statement I is incorrect but Statement II is correct.
2.	Both Statement I and Statement II are correct.
3.	Both Statement I and Statement II are incorrect.
4.	Statement I is correct but Statement II is incorrect.

Q17. Match List - I with List - II. **NEET - 2022**

	List-I (Products formed)		List - II (Reaction of carbonyl compound with)
(A)	Cyanohydrin	(I)	NH ₂ OH
(B)	Acetal	(II)	RNH ₂
(C)	Schiff's base	(III)	Alcohol
(D)	Oxime	(IV)	HCN

Choose the correct answer from the options given below:



	(A)	(B)	(C)	(D)
1.	(IV)	(III)	(II)	(I)
2.	(III)	(IV)	(II)	(I)
3.	(II)	(III)	(IV)	(I)
4.	(I)	(III)	(II)	(IV)

Q18.

NEET - 2022

$\text{RMgX} + \text{CO}_2 \xrightarrow{\text{dry}} \text{ether } \text{Y} \xrightarrow{\text{H}_3\text{O}^+} \text{RCOOHRMgX} + \text{CO}_2 \xrightarrow{\text{dry ether}} \text{Y} \xrightarrow{\text{H}_3\text{O}^+} \text{RCOOH}$

What is Y in the above reaction?

1. $(\text{RCOO})_2\text{Mg}$
2. $\text{RCOO}^-\text{Mg}^+\text{X}$
3. $\text{R}_3\text{CO}^-\text{Mg}^+\text{X}$
4. RCOO^-X^+

Q19. Which one of the following is not formed when acetone reacts with 2-pentanone in the presence of dilute NaOH followed by heating?

NEET - 2022

1.		2.	
3.		4.	

Q20. Given below are two statements:

NEET - 2022

Statement I:	Aldehydes and ketones having at least one α -hydrogen undergo aldol condensation in the presence of dilute alkali as catalyst.
Statement II:	When aldol condensation is carried out between two different aldehydes, it is called cross aldol condensation. Ketones do not give this reaction.

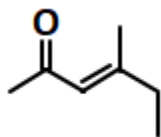


In light of the above statements, choose the most appropriate answer from the options given below:

1.	Statement I is correct and statement II is incorrect.
2.	Statement I is incorrect and statement II is correct.
3.	Both statement I and statement II are correct.
4.	Both statement I and statement II are incorrect.

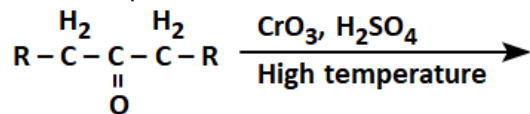


Q21. The correct IUPAC name of the following compound is: **NEET - 2022**



1.	2-ethylhex-3-en-4-one	2.	4-methylhex-3-en-2-one
3.	4-ethylpent-3-en-2-one	4.	3-methylhex -3-en-4-one

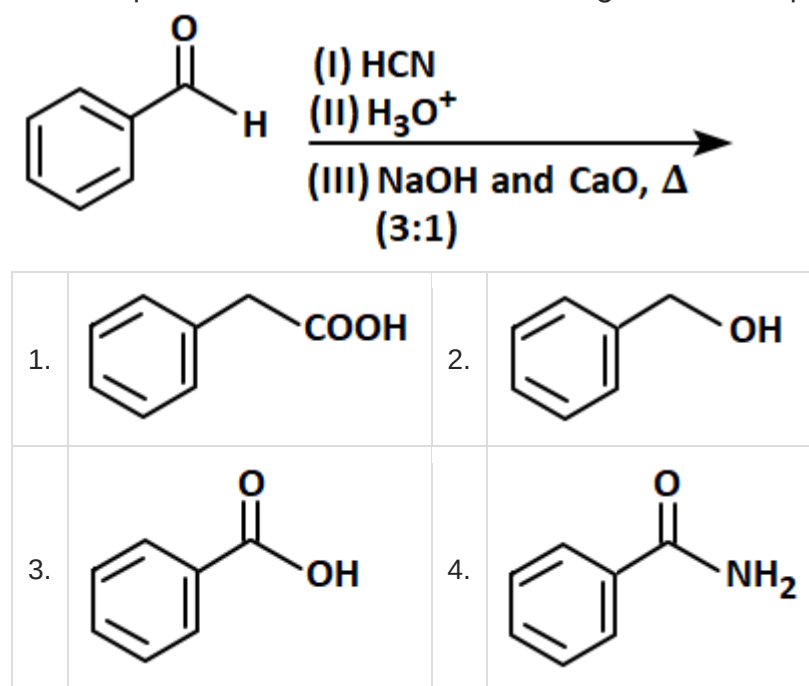
Q22. The product(s) formed from the following reaction is/are: **NEET - 2022**



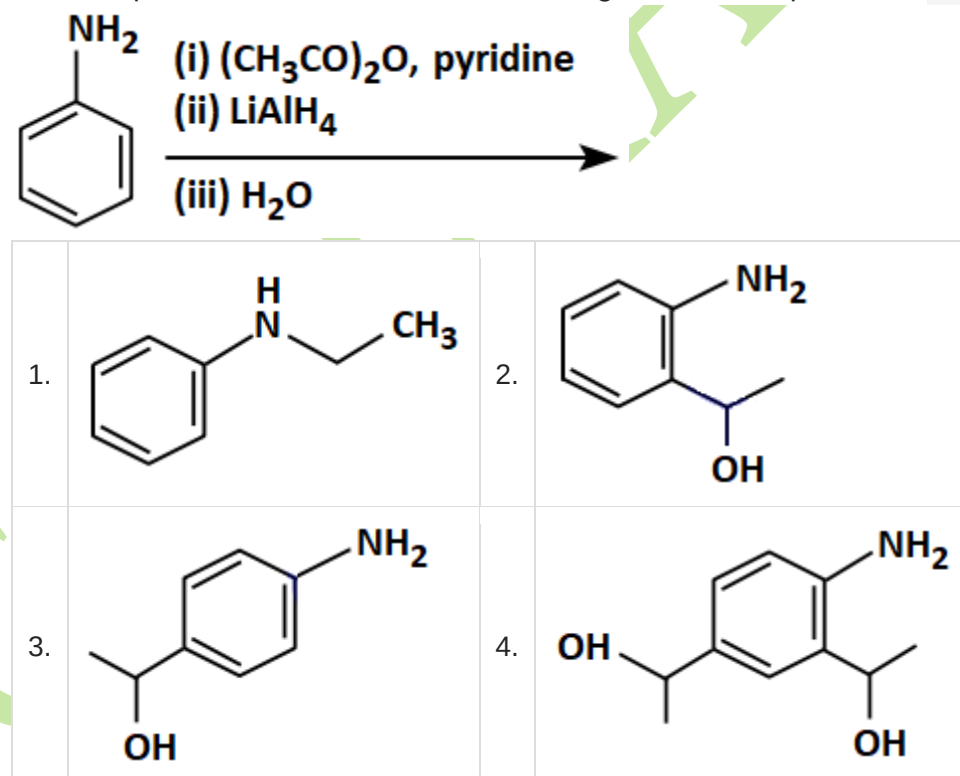
1.	RCOOH only
2.	RCH ₂ COOH only
3.	$\begin{array}{c} \text{H}_2 \quad \text{H} \quad \text{H}_2 \\ \text{R}-\text{C}-\text{C}-\text{C}-\text{R} \\ \\ \text{COOH} \end{array} \quad \text{only}$
4.	RCOOH and RCH ₂ COOH



Q23. The product formed from the following reaction sequence is: NEET - 2022



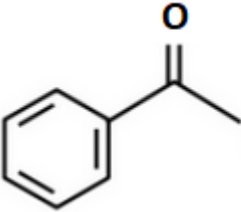
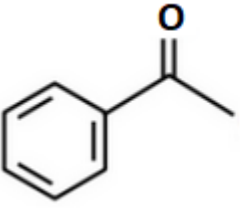
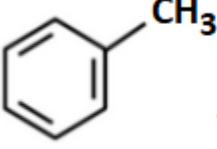
Q24. The product formed from the following reaction sequence is: NEET - 2022



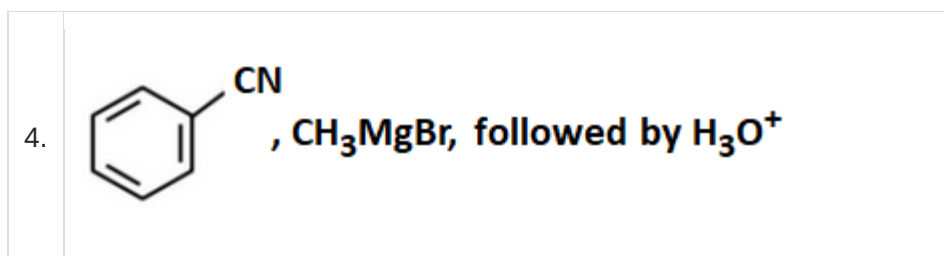
Q25. Which of the following reactions is not an example of nucleophilic addition-elimination reaction? NEET - 2022

1.	$\text{CH}_3\text{CHO} + \text{NaHSO}_3 \rightleftharpoons \text{H}_3\text{C} - \overset{\text{OH}}{\underset{\text{H}}{\text{C}}} - \text{OSO}_2\text{Na}$
2.	$\text{CH}_3\text{CHO} + \text{NH}_2\text{OH} \rightleftharpoons \text{CH}_3\text{CH} = \text{N} - \text{OH} + \text{H}_2\text{O}$
3.	$\text{CH}_3\text{CHO} + \text{C}_6\text{H}_5\text{NHNH}_2 \rightleftharpoons \text{CH}_3\text{CH} = \text{N} - \text{NHC}_6\text{H}_5 + \text{H}_2\text{O}$
4.	$\text{CH}_3\text{CHO} + \text{NH}_3 \rightleftharpoons \text{CH}_3\text{CH} = \text{NH} + \text{H}_2\text{O}$

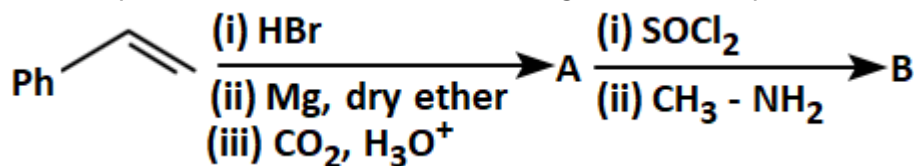
Q26. The incorrect method to synthesize benzaldehyde is: NEET - 2022

1.	 $\text{Cl}, \text{H}_2, \text{Pd-BaSO}_4$
2.	 $\text{OC}_2\text{H}_5, \text{DIBAL-H, followed by H}_2\text{O}$
3.	 $, \text{CrO}_2\text{Cl}_2, \text{ followed by H}_3\text{O}^+ \text{ in CS}_2$



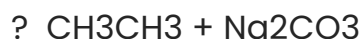


Q27. The products A and B in the following reaction sequence are : NEET - 2022



1.	$\text{A} = \text{Ph}-\text{CH}_2-\text{CH}_2-\text{COOH}$; $\text{B} = \text{Ph}-\text{CH}_2-\text{CH}_2-\text{CONHCH}_3$
2.	$\text{A} = \text{Ph}-\text{CH}(\text{CH}_3)-\text{COOH}$; $\text{B} = \text{Ph}-\text{CH}(\text{CH}_3)-\text{CONHCH}_3$
3.	$\text{A} = \text{Ph}-\text{CH}_2-\text{CH}_2-\text{OH}$; $\text{B} = \text{Ph}-\text{CH}_2-\text{CH}_2-\text{NHCH}_3$
4.	$\text{A} = \text{Ph}-\text{CH}(\text{CH}_3)-\text{OH}$; $\text{B} = \text{Ph}-\text{CH}(\text{CH}_3)-\text{NHCH}_3$

Q28. Consider the below reaction and identify the missing reagent/chemical. NEET - 2021



1. CaO

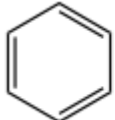
2. DIBAL-H



3. B_2H_6

4. Red Phosphorus

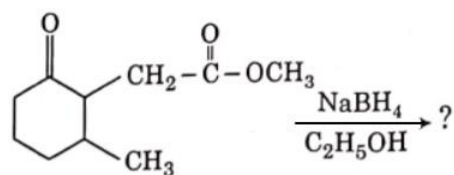
Q29. Match List-I with List-II. NEET - 2021

	List-I (Substrates of reactions)		List II (Name of reactions)
(a).	 $\xrightarrow[\text{Anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$	(i).	Hell-Volhard-Zelinsky reaction
(b).	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX}$	(ii).	Gatterman n-Koch reaction
(c).	$\text{R}-\text{CH}_2-\text{OH} + \text{RCOOH} \xrightarrow[\text{H}_2\text{SO}_4]{\text{Conc. H}_2\text{SO}_4} \text{R}-\text{CH}_2-\text{OH} + \text{RCOOH} \rightarrow \text{Conc.}$	(iii).	Haloform reaction
(d).	$\text{R}-\overset{\text{H}_2}{\text{C}}-\text{COOH} \xrightarrow[2. \text{H}_2\text{O}]{1. \text{X}_2/\text{Red P}}$	(iv).	Esterification

Choose the correct answer from the options given below:

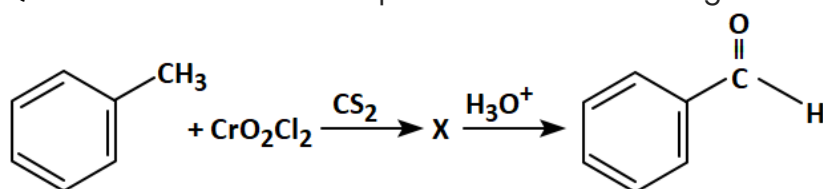
	(a).	(b).	(c).	(d).
1.	(i).	(iv).	(iii).	(ii).
2.	(ii).	(iii).	(iv).	(i).
3.	(iv).	(i).	(ii).	(iii).
4.	(iii).	(ii).	(i).	(iv).

Q30. The product formed in the following chemical reaction is : NEET - 2021



1.		2.	
3.		4.	

Q31. The intermediate compound 'X' in the following chemical reaction is : **NEET - 2021**



1.	
2.	
3.	
4.	

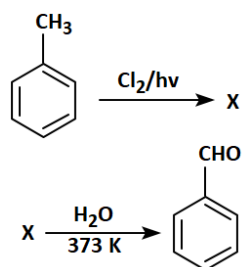
Q32. The reaction between benzaldehyde and acetophenone in the presence of dilute NaOH is known as: **NEET - 2020**



1. Cannizzaro's reaction
2. Cross Cannizzaro's reaction
3. Cross aldol condensation
4. Aldol condensation

Q33. The product 'X' in the below mentioned reaction is:

NEET - 2020



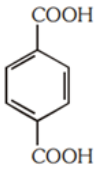
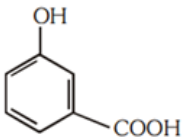
1.		2.	
3.		4.	

Q34. An acid that forms an anhydride (X) on heating and an acid imide (Y) on strong heating with ammonia is:

NEET - 2021

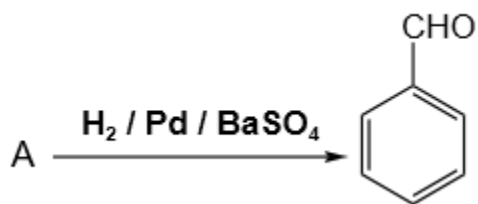
1.		2.	
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3.		4.	
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Q35. The reactant 'A' in the below-mentioned reaction is:

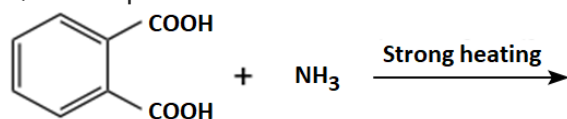
NEET - 2021

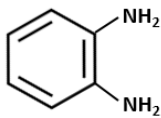
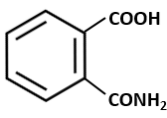
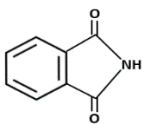
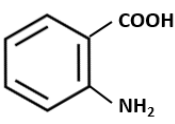


1. Benzoyl chloride	2. Toluene
3. Acetophenone	4. Benzoic acid

Q36. The product of the below-mentioned reaction is:

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1. 	2. 
3. 	4. 



ANSWER KEY

1. Ans. (2)
2. Ans. (4)
3. Ans. (3)
4. Ans. (3)
5. Ans. (2)
6. Ans. (3)
7. Ans. (2)
8. Ans. (3)
9. Ans. (1)
10. Ans. (2)
11. Ans. (4)
12. Ans. (2)
13. Ans. (4)
14. Ans. (4)
15. Ans. (2)
16. Ans. (2)
17. Ans. (1)
18. Ans. (2)
19. Ans. (3)
20. Ans. (1)
21. Ans. (2)
22. Ans. (4)
23. Ans. (2)
24. Ans. (1)
25. Ans. (1)
26. Ans. (4)
27. Ans. (2)
28. Ans. (1)
29. Ans. (2)
30. Ans. (2)
31. Ans. (3)
32. Ans. (3)
33. Ans. (2)
34. Ans. (1)
35. Ans. (1)
36. Ans. (3)

